INCH-POUND
MIL-H-24729(SH)
26 January 1990

MILITARY SPECIFICATION

HARNESS, SAFETY, SWIMMERS

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification is for an adjustable, nylon-web harness which allows freedom of movement by swimmers or personnel engaged in other hazardous operations where the attachment of a line is desirable.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications, and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

V-T-295	Thread, Nylon
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-F-320	Fiberboard: Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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MILITARY

MIL-W-4088	Webbing, Textile, Woven Nylon
MIL-C-5040	Cord, Fibrous, Nylon
MIL-L-19140	Lumber and Plywood, Fire-Retardant Treated
MIL-F-21840	Fastener Tapes, Hook and Pile, Synthetic
MS 22045	Ring-Parachute Harness "V"
MS 70101	Adapter, Reversible, Quick Fit, 500 Lb; 1-3/4 Inch
MS 70102	Ring, Parachute Harness, Quick-Fit, "V" Type

STANDARDS

FEDERAL

FED-STD-751	Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-2073-1	DOD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government drawing. The following other Government drawing forms a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

803-6397300	Harness, Safety, Swimmers-Assembly and Details

(Application for copies should be addressed to Commander, Portsmouth Naval Shipyard, Code 202.2, Portsmouth, NH 03801.)

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2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 117 Standard Method of Salt Spray (Fog) Testing (DOD adopted)
D 3951 Standard Practice for Commercial Packaging (DOD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2) a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Materials.

3.2.1 Recovered materials. Unless otherwise specified herein, all material and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specified.

3.3 Construction. The design, manufacturing, and assembly details of the swimmers safety harness shall conform to Drawing 803-6397300.

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3.3.1 Webbing. The webbing for the harness shall be a nominal 1-3/4 inch wide nylon and shall be type XXVII. The chemical light pocket shall consist of both type IV, 3 inch wide nylon webbing and type II, 1 inch wide nylon webbing, as required. The parachute shroud cutter J-knife pocket shall consist of both type VIII, a nominal 1-3/4 inch wide nylon webbing and type II, 1 inch wide nylon webbing, as required. Webbing used in the harness shall conform to MIL-W-4088.

3.3.2 Webbing ends. Webbing ends shall be seared to prevent fraying.

3.3.3 Color. The types XXVII, VIII and IV webbing shall closely match color olive drab 7 from MIL-W-4088. The color of the type II webbing shall closely match the Air Force yellow 1365 from MIL-W-4088.

3.3.4 Thread. The thread used shall conform to V-T-295 and shall be size E or number 6, as applicable. Color shall match the harness as closely as possible.

3.3.5 Stitches and stitching. Stitching shall conform to FED-STD-751 and shall be as specified in Drawing 803-6397300.

3.3.6 Buckles. Buckles shall meet their minimum proof loads, as specified in MS 22045, MS 70101, or MS 70102, as applicable.

3.3.7 Lanyards. The lanyards used to attach both the J-knife and the strobe light to the harness shall each be 48 inches long and shall conform to type I of MIL-C-5040.

3.3.8 Tape fasteners. The nylon tape fasteners used to secure the belt ends and pocket flaps shall be type IA, class 1 color 106 of MIL-F-21840, as specified in 3.3.3. The hooks and pile shall be attached to the belt and flaps in such a manner and of sufficient length as to form mating surfaces as detailed on Drawing 803-6397300.

3.3.9 Pockets. The pocket holding the SDU-5/E strobe light shall be constructed from the pocket supplied as part of the light assembly and as modified on Drawing 803-6397300. The pockets for the chemical lights shall be constructed to hold both 6 inch long wands and 4 inch long wands as detailed on Drawing 803-6397300. The pocket for holding the J-knife shall be constructed as detailed on Drawing 803-6397300. Pocket flaps shall have yellow nylon pull tabs attached to facilitate ease of opening of pockets. Materials used shall be as specified in 3.2 and 3.3.

3.4 Performance.

3.4.1 Salt spray. The harness shall show no signs of loosing, fraying or separation of fiber and the hardware shall show no evidence of corrosion (see 4.5.2).

3.4.2 Static load. The harness shall show no signs of loosening, fraying or separation of fiber (see 4.5.3).

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3.4.3 V-ring belt closure. The V-ring belt closure shall show no signs of slippage or distortion (see 4.5.4).

3.4.4 Reversible adapter closure. The reversible adapter closure shall show no signs of slippage or distortion (see 4.5.5).

3.4.5 V-ring attachment. The V-ring attachment shall show no signs of loosening, fraying or separation of fiber and the hardware shall show no signs of slippage or distortion (see 4.5.6).

3.4.6 Hardware. Hardware shall show no signs of slippage or distortion (see 4.5.7).

3.5 Marking. The marking shall be permanent and legible.

3.5.1 Labelling. A water resistant label shall be sewn on the inside of the chest strap between the back strap and the left riser strap and shall include the following:

(a) Harness, Safety, Swimmers
(b) MIL-H-24729
(c) Contract number ----------
(d) Date of manufacture ----------
(e) Date put in service ---------- (leave blank).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination and tests specified in 4.5 (see 6.3). Failure of any examination or test shall constitute failure of the first article inspection.

4.3.1 Sampling for first article inspection. Prior to beginning production, two sample harnesses produced with equipment and procedures used in production shall be subjected to the first article inspection (see 6.4).

4.4 Quality conformance inspection. Quality conformance inspection shall consist of group A and group B inspections as specified in 4.4.2 and 4.4.3 (see 6.3).

4.4.1 Inspection lot. An inspection lot shall consist of all harnesses produced under essentially the same conditions and offered for inspection at one time.

4.4.2 Group A inspection. Every item in each lot shall be examined as specified in 4.5.1. Classification of defects shall be as shown in table I.

TABLE I. *Classification of defects.*

Categories	Defects
Critical:	
1	Webbing frayed, ragged, or loose strands
2	Buckle parts not mating, locking or unlocking with facility
3	Intertwined, tangled, bunched, or broken threads in seams or stitching
Major:	
101	Incomplete; any component parts missing
102	Material not as specified
103	Pockets missing, incomplete or not as specified
104	Parts or size not as specified
105	Metal parts not free of burrs and sharp edges
106	Stitches not firmly in place, not evenly spaced
107	Not proper number of stitches per inch
108	Color not as specified or uneven shade
Minor:	
201	Thread ends not snipped
202	Marking missing, incorrect, illegible, not permanent, or not located as specified

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4.4.2.1 Failures. One or more discrepancies falling within the critical or major categories of table I shall be cause for rejection of the lot. Minor discrepancies may be repaired and resubmitted for inspection.

4.4.3 Group B inspection. A random sample of harnesses shall be selected from each lot in accordance with table II. The samples shall be subjected to the tests of 4.5.3 through 4.5.6.

TABLE II. *Sampling plan*[1,2,3]

Lot size	Sample size	Accept	Reject
0-50	2	0	1
51-500	3	0	1
501-35,000	5	0	1
over 35,000	8	0	1

[1]All defective items must be replaced with acceptable items prior to lot acceptance.
[2]Inspect sample size until the reject criteria is reached. If reject criteria is reached, reject the lot or inspect the entire lot.
[3]If entire sample is inspected and reject criteria is not reached, accept the lot.

4.4.3.1 Failures. If any sample unit fails to pass group B inspection, the lot shall be rejected.

4.4.3.2 Disposition of sample units. Sample units which have been subjected and passed group B inspection shall be delivered on the contract.

4.5 Examination and tests. Examination and tests shall be as specified in 4.5.1 through 4.5.7 (see 6.3).

4.5.1 Visual and mechanical examination. Safety harnesses shall be examined to verify that the materials, design, construction, physical dimensions, marking and workmanship are in accordance with the applicable requirements (see 3.2 through 3.5).

4.5.2 Salt spray. The safety harness shall be subjected to 96 hours of salt spray (fog) testing in accordance with ASTM B 117 prior to load tests of 4.5.3 through 4.5.6, as applicable. The hardware shall be examined for evidence of excessive corrosion (see 6.5). The safety harness material shall conform to the requirements of 3.3 and 3.4.1.

4.5.3 Static load. The load tests shall be performed using a standard tensile testing machine. The force shall be applied at a rate of 2.0 plus or minus 1 inch per minute until the values specified herein are achieved. The load scale shall be accurate to plus or minus 5 percent (see 4.5.7).

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4.5.4 V-ring belt closure. The belt shall be closed as in a normal use procedure. The belt closure V-ring as shown on MS 70102 shall be placed between the jaws of the testing machine so that approximately 10 inches of the belt has been clamped for testing. The rate of separation of the jaws of the testing machine shall be approximately 2 inches per minute. The load shall be increased to 360 pounds and held at that value for 1 minute (see 4.5.7).

4.5.5 Reversible adapter closure. The reversible adapter closure as shown on MS 70101 shall be placed between the jaws of the testing machine so that the adapter is at the midpoint between the jaws. The rate of separation of the jaws of the testing machine shall be approximately 2 inches per minute. The load shall be increased to 360 pounds and held at the value for 1 minute (see 4.5.7).

4.5.6 V-ring attachment. A section of webbing shall be attached to the base of the V-ring. To the apex of the ring there shall be attached a cable or other suitable non-stretchable attachment. The V-ring shall be as close to the midpoint of the separated jaws as possible. The rate of separation shall be approximately 2 inches per minute. The load shall be increased to 360 pounds and held at that value for 1 minute (see 4.5.7).

4.5.7 Hardware. Prior to attachment to webbing straps, random samples of all hardware will be selected by the Government inspector from each lot offered for inspection in accordance with procedures of MIL-STD-105, inspection level I. Samples shall meet the minimum requirements specified in MS 70101, MS 70102 and MS 22045.

4.6 Inspection of packaging. Sample packs, and the inspection of the preservation, packing and marking for shipment, stowage and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging or preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 General.

5.1.1 Navy fire-retardant requirements.

(a) *Treated lumber and plywood.* When specified (see 6.2), all lumber and plywood including laminated veneer material used in shipping container and pallet construction, members, blocking, bracing, and reinforcing shall be fire-retardant treated material conforming to MIL-L-19140 as follows:

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Levels A and B	Type II - weather resistant Category I - general use
Level C	Type I - non-weather resistant Category I - general use.

(b) *Fiberboard.* Fiberboard used in the construction of class-domestic, non-weather resistant fiberboard and cleated fiberboard boxes including interior packing forms shall meet the flamespread index and the specific optic density requirements of PPP-F-320.

5.2 Preservation. Preservation shall be level A, C or commercial as specified (see 6.2).

5.2.1 Level A. Each harness shall be folded in accordance with the manufacturer's practice to a minimum practical size without sustaining harness damage during stowage and storage. The folded harness shall then be enclosed in a transparent, flexible, plastic bag of minimum 0.004 mil thickness. Bag closure shall be of the press-fit type or by heat sealing. Each harness shall then be placed in a fiberboard box conforming to PPP-B-636, class-weather resistant. The box type, variety and style shall be at the supplier's option. Box closure shall be in accordance with method V of the appendix to the box specification.

5.2.2 Level C. Harnesses shall be unit protected as specified for level A except, and unless otherwise specified (see 6.2), the fiberboard box shall be of the class-domestic/fire-retardant type, and closure shall be in accordance with method I using pressure-sensitive adhesive tape.

5.2.3 Commercial. Commercial packaging (cleanliness, preservation and unit pack) shall be in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, B, C, or commercial as specified (see 6.2).

5.3.1 General requirements for levels A, B, and C. Containers selected (see 5.3.2) shall be of minimum weight and cube consistent with the protection required, and of uniform size.

5.3.2 Levels A, B, and C containers. Harnesses preserved as specified (see 5.2), shall be packed in exterior shipping containers for the level of packing specified (see 5.3), in accordance with table VII, exterior shipping container requirements, of MIL-STD-2073-1, appendix C, and herein. Unless otherwise specified (see 6.2), container selection and options shall be at the contractor's option.

5.3.2.1 Caseliners, closure and gross weight.

5.3.2.1.1 Caseliners. Unless otherwise specified (see 6.2), level A shipping containers containing harnesses preserved level C or commercial shall be provided with waterproof caseliners in accor-dance with MIL-STD-2073-1.

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5.3.2.1.2 Closure. Container closure, reinforcing, or banding shall be in accordance with the applicable container specification or appendix thereto except that weather-resistant fiberboard boxes shall be closed in accordance with method V and reinforced with non-metallic or tape banding and domestic or fire-retardant fiberboard boxes shall be closed in accordance with method I using pressure sensitive tape.

5.3.2.1.3 Weight. Wood, plywood, and cleated type containers exceeding 200 pounds gross weight shall be modified by the addition of skids in accordance with MIL-STD-2073-1 and the applicable container specification or appendix thereto.

5.3.3 Commercial. Harnesses preserved as specified (see 5.2), shall be packed for shipment in accordance with ASTM D 3951 and herein.

5.3.3.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage and storage.

5.4 Marking, Levels A, B, C, and commercial. In addition to any special marking required (see 6.2 and 3.5), interior (unit) packs and shipping containers shall be marked including bar coding for shipment, stowage, and storage in accordance with MIL-STD-2073-1, appendix F.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Harnesses covered by this specification are intended for use by swimmers engaged in water rescue operations where freedom of arm movement and attachment of a safety line is desirable. The service life of this harness is 7 years from the date placed in service or 8 1/2 years from the date of manufacture, whichever occurs first. When the harness reaches its service life limit, it will be removed from service and scrapped.

6.2. Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)
(c) When first article is required (see 3.1)
(d) When fire-retardant material is required (see 5.1.1(a))
(e) Level of preservation and packing required (see 5.2 and 5.3)

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(f) When a fire-retardant fiberboard box is not required (see 5.2.2)
(g) Container selection and options if other than contractor's option (see 5.3.2)
(h) When waterproof caseliners are not required (see 5.3.2.1.1)
(i) Special making required (see 5.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3	DI-T-4902	First article inspection report	---
4.4	DI-T-5329	Inspection and test reports	---
4.5	DI-R-7065	Equipment calibration procedures	---

The above DIDs were those cleared as of the date of this standard. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

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6.4.1 First article samples. If the Government requires that first article tests are to be conducted at a laboratory, the following material should be submitted, by the Government inspector, to the David Taylor Research Center, Code 2843, Annapolis, MD 21402-5067; marked with the appropriate classification as noted in 4.2:

(a) Two complete harnesses
(b) Four sets of all hardware used in the harness
(c) One cone of all thread used in the harness
(d) Five yards each of all webbing, tapes and lanyard line used in the harness.

6.5 Definitions.

6.5.1 Excessive corrosion. Excessive corrosion is that which interferes with operation or, in the case of plate metals, corrosion which has passed through the plating and attacked the base metal.

6.6 Sub-contracted material and parts. The packaging or preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 The Acceptable Quality Level (AQL) for the inspection of hardware specified in 4.5.7 should be 1.0 (see table IV-A of MIL-STD-105).

6.8 Subject term (key-word) listing.

Lanyards
Pockets
Tape fasteners
V-ring attachment
V-ring belt closure
Webbing

Preparing activity:
Navy - SH
(Project4240-N949)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-24729(SH)	HARNESS, SAFETY, SWIMMERS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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